

SIH26139: Detailed Data Preprocessing & Pipeline Architecture

This document provides a block-by-block explanation of the `prepare_data.py` script, detailing how raw medical data is transformed into quantum-ready matrices.

Module 1: Secure Data Acquisition (`download_all`)

What it does: Fetches raw CSV files directly from verified public mirrors.

How it works: Validates the integrity of the downloaded data by comparing the resulting DataFrame's shape (rows/cols) against the officially documented shape. It then generates a SHA-256 digital hash.

Presentation Pitch: 'We automated data ingestion with SHA-256 hashing to prove to the jury that zero synthetic or tampered data was used.'

Module 2: Data Cleaning & Hygiene

What it does: Removes logical errors and prevents data leakage.

How it works: For the Heart track, it drops 272 duplicate rows caused by a historical database merge. For the Pima dataset, it replaces biologically impossible zeros (e.g., blood pressure = 0) with NaN. For the BRFSS dataset, it stratifies and subsamples 253,680 rows down to 100,000 to optimize training.

Presentation Pitch: 'We applied strict medical data hygiene—dropping duplicates and handling biological impossibilities—followed by strict train/test splitting to guarantee zero data leakage.'

Module 3: Ensemble Feature Selection

What it does: Mathematically selects the most predictive clinical features while dropping redundant ones.

How it works: Runs a '6-Council' test (Mutual Info, RF Importance, XGBoost Gain, Permutation, SHAP, and CV Stability). It averages the ranks. It then calculates the Spearman correlation matrix and drops features that are >90% correlated with stronger features (Redundancy Pruning).

Presentation Pitch: 'Rather than guessing, we mathematically ranked features using 6 different algorithms, pruning redundant data to isolate the strongest clinical signals.'

Module 4: Classical Scaling & Imputation

What it does: Standardizes the numbers so classical machine learning models treat all features equally.

How it works: Uses *SimpleImputer* (median strategy) to fill missing values, and *StandardScaler* to map data to a mean of 0 and std-dev of 1. Crucially, these are fitted ONLY on the training split.

Presentation Pitch: 'We applied median imputation and Z-score scaling fitted strictly on training data, ensuring the model treats Age and Cholesterol with equal mathematical weight without peaking at the test set.'

Module 5: Quantum Encoding Matrix Generation

What it does: Translates standard 2D data into quantum state formats (physical angles and unit vectors).

How it works:

1. **Angle Encoding:** Uses MinMaxScaler to map features into the $[0, \pi]$ range to act as physical qubit rotation angles.
2. **Amplitude Encoding:** Because amplitude vectors require dimensions to be powers of 2, the pipeline zero-pads the data (e.g., 6 features padded to 8 dims) to fit on 3 qubits (2^3). It then applies L2-normalization so the vector magnitude equals 1.0 (a law of quantum mechanics).

Presentation Pitch: 'Quantum simulators cannot read CSVs. Our pipeline automatically zero-pads and L2-normalizes the clinical data into a Hilbert space representation for PennyLane VQC ingestion.'

Module 6: The Audit Trail (metadata.json)

What it does: Generates a massive JSON receipt of the entire pipeline execution.

How it works: Logs original row counts, split ratios, exact dropped features, and SHA-256 hashes.

Presentation Pitch: 'We built an automated audit trail to guarantee complete transparency and reproducibility for our medical pipeline.'